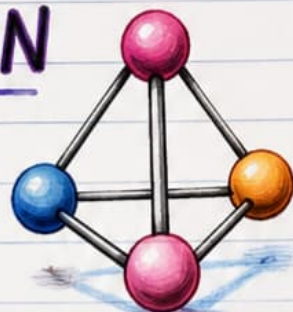


SYSTEM OF PARTICLES

AND

ROTATIONAL MOTION

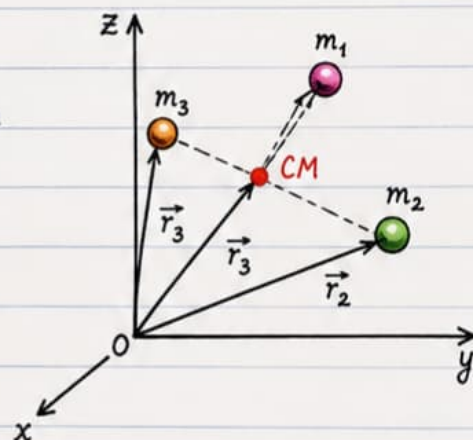
When **two or more particles** interact with each other, the system is called a **System of Particles**. If the particles of a system do not change their relative positions with time, the system is called a **Rigid Body**.



1. CENTRE OF MASS (CM)

- It is the point where the entire mass of the system may be considered to be concentrated.
- It is denoted by 'CM' or 'G'.
- Position vector of CM of a system of particles:

$$\vec{R}_{cm} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots + m_n \vec{r}_n}{m_1 + m_2 + \dots + m_n}$$



2. MOMENTUM OF SYSTEM

- Total momentum of a system = Vector sum of momenta of all particles.

$$\vec{P} = \vec{p}_1 + \vec{p}_2 + \dots + \vec{p}_n$$

IMPORTANT POINT

If no external force acts on the system, total momentum of the system remains constant.

3. CONSERVATION OF LINEAR MOMENTUM

- If resultant external force on the system is zero, the total linear momentum of the system remains constant in magnitude and direction.

$$\vec{P}_{initial} = \vec{P}_{final}$$

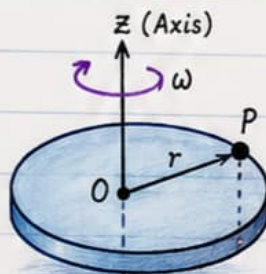
EXAMPLE

Two particles m_1 and m_2 moving with velocities $\vec{u}_1$ and $\vec{u}_2$ collide and move with velocities $\vec{v}_1$ and $\vec{v}_2$. If no external force acts,

$$m_1 \vec{u}_1 + m_2 \vec{u}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

4. ROTATIONAL MOTION

- When a rigid body rotates about a **fixed axis**, each particle of the body describes a **circular path** about that axis.
- Rotational motion is characterized by **angular quantities**.



Key Terms

- Axis of rotation
- Angular displacement (θ)
- Angular velocity (ω)
- Angular acceleration (α)

5. MOMENT OF A FORCE (TORQUE)

- When a force acts on a body, it may rotate the body about a fixed axis.
- The turning effect of a force about an axis is called **Moment of the Force** or **Torque**.

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Where,

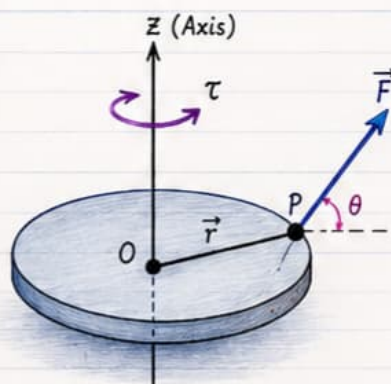
$\vec{r} \rightarrow$ position vector of point of application of force w.r.t. axis.

$\vec{F} \rightarrow$ applied force

- Magnitude of torque :

$$\tau = r F \sin \theta$$

$\theta =$ angle between $\vec{r}$ and $\vec{F}$



Direction of $\vec{\tau}$ is given by Right Hand Thumb Rule.

IMPORTANT

- If $\theta = 0^\circ$ or 180° then $\vec{\tau} = 0$ (No rotation)
- If $\theta = 90^\circ$ then $\vec{\tau}$ is maximum.

Unit of Torque

$\rightarrow$ N-m (Newton-metre)
(SI unit)

6. MOMENT OF INERTIA (I)

- It is the property of a body by virtue of which it **opposes any change in its state of rotation**.
- For a system of particles about a given axis,

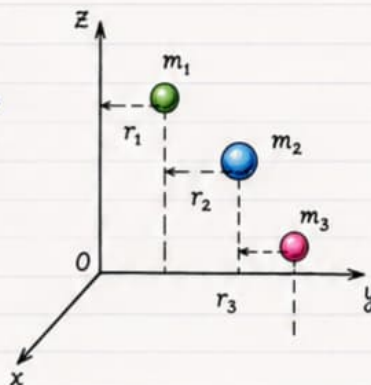
$$I = \sum m_i r_i^2$$

Where,

$m_i =$ mass of i^{th} particle

$r_i =$ perpendicular distance of i^{th} particle from the axis

$I =$ moment of inertia



Special Cases

- Point mass at distance r :

$$I = mr^2$$

- Two point masses m_1, m_2 at r_1, r_2 :

$$I = m_1 r_1^2 + m_2 r_2^2$$

- n point masses :

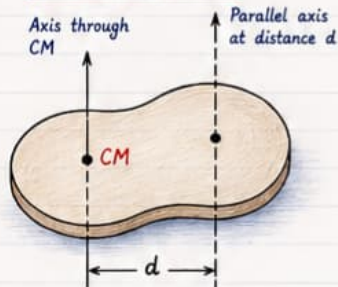
$$I = \sum_{i=1}^n m_i r_i^2$$

7. THEOREMS FOR MOMENT OF INERTIA

(a) Parallel Axis Theorem

The moment of inertia about any axis parallel to a given axis is equal to the moment of inertia about the given axis + product of mass of the body and square of distance (d) between the two axes.

$$I_P = I_{CM} + Md^2$$



EXAMPLE

A uniform disc of mass M and radius R .

About central axis :

$$I_{CM} = \frac{1}{2} MR^2$$

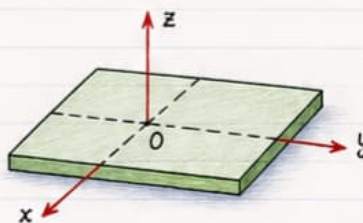
About a parallel axis at distance d :

$$I_P = \frac{1}{2} MR^2 + Md^2$$

(b) Perpendicular Axis Theorem

For a plane lamina, the moment of inertia about an axis perpendicular to its plane and passing through a point is equal to the sum of the moments of inertia about two mutually perpendicular axes in its plane passing through the same point.

$$I_z = I_x + I_y$$



EXAMPLE

For rectangular lamina (length = a , breadth = b , mass = M)

$$I_x = \frac{1}{12} Mb^2$$

$$I_y = \frac{1}{12} Ma^2$$

$$\Rightarrow I_z = \frac{1}{12} M(a^2 + b^2)$$

QUICK SUMMARY

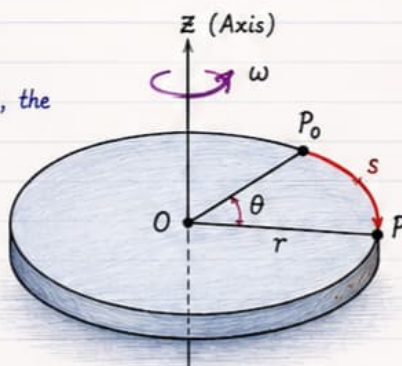
- ✓ Rigid body $\rightarrow$ System of particles with fixed relative positions.
- ✓ CM $\rightarrow$ Point representing whole mass.
- ✓ Momentum $\rightarrow$ Vector sum of individual momenta.
- ✓ Torque $\rightarrow$ Turning effect of a force ($\vec{\tau} = \vec{r} \times \vec{F}$).
- ✓ Moment of inertia $\rightarrow I = \sum m_i r_i^2$ (depends on mass distribution and axis).
- ✓ Parallel axis theorem $\rightarrow I_P = I_{CM} + Md^2$
- ✓ Perpendicular axis theorem $\rightarrow I_z = I_x + I_y$ (for plane lamina).

Remember!

- Axis matters in rotational motion.
- Same body can have different moment of inertia about different axes.
- Greater moment of inertia $\rightarrow$ more resistance to change in rotation.

8. ANGULAR DISPLACEMENT (θ)

- When a body rotates about a fixed axis, the angle through which it turns is called **angular displacement**.
- It is a vector quantity.
- Unit : radian (rad)

Relation

If a point on the body moves through an arc length s on a circle of radius r , then

$$\theta = \frac{s}{r}$$

(for small angles)

9. ANGULAR VELOCITY (ω)

- The time rate of change of angular displacement is called **angular velocity**.
- It is a vector quantity.

$$\omega = \frac{d\theta}{dt}$$

Unit : rad s^{-1}

Important Points

- ω gives the speed of rotation.
- Direction of $\vec{\omega}$ is along the axis of rotation (Right Hand Thumb Rule).
- For uniform rotation, $\vec{\omega}$ is constant.



R.H.T Rule

10. ANGULAR ACCELERATION (α)

- The time rate of change of angular velocity is called **angular acceleration**.
- It is a vector quantity.

$$\alpha = \frac{d\omega}{dt}$$

Unit : rad s^{-2}

Types of Rotation

Type	ω	α
Uniform Rotation	Constant	Zero
Non-uniform Rotation	Changing	Non-zero
Uniformly Accelerated Rotation	Changing uniformly	Constant

Kinematic Equations for Uniformly Accelerated Rotation

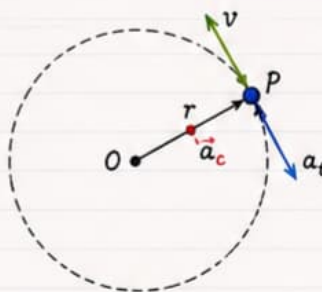
- $\omega = \omega_0 + \alpha t$
- $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$
- $\omega^2 = \omega_0^2 + 2\alpha\theta$

Where, ω_0 = initial angular velocity
 ω = final angular velocity
 α = angular acceleration
 t = time, θ = angular displacement

11. RELATION BETWEEN LINEAR AND ANGULAR QUANTITIES

For a particle at distance r from the axis :

Linear Quantity	Angular Quantity	Relation
Linear Displacement (s)	Angular Displacement (θ)	$s = r\theta$
Linear Velocity (v)	Angular Velocity (ω)	$v = r\omega$
Linear Acceleration (a_t)	Angular Acceleration (α)	$a_t = r\alpha$
Centripetal Acceleration (a_c)	—	$a_c = \frac{v^2}{r} = r\omega^2$

Note :

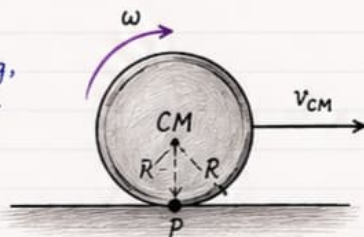
a_t = tangential acceleration (changes speed)
 a_c = centripetal acceleration (changes direction)

12. ROLLING WITHOUT SLIPPING

- When a body rolls on a surface without slipping, the point of contact is instantaneously at rest.
- Relation between linear and angular velocity :

$$v_{CM} = R\omega$$

Where, R = radius of the body

Example

A solid cylinder of radius R rolls without slipping with angular velocity ω . Then, velocity of its centre of mass is

$$v_{CM} = R\omega$$

SUMMARY

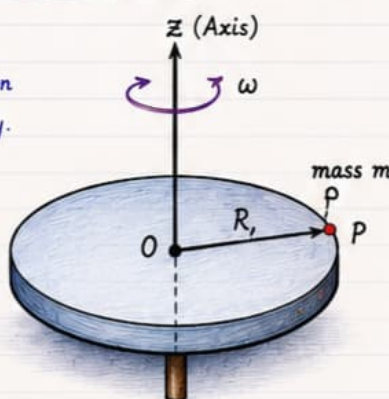
- System of particles $\rightarrow$ Treated as a single entity.
- CM represents the whole system.
- Momentum = Vector sum of individual momenta.
- Torque = Turning effect of a force about an axis.
- Moment of inertia $\rightarrow$ Resistance to change in rotational motion.
- Rotational motion described by θ, ω, α .
- Linear and angular quantities are related.

13. ROTATIONAL KINETIC ENERGY (K.E.)

- The kinetic energy of a body in pure rotational motion depends on its moment of inertia and angular velocity.
- For a rigid body rotating about a fixed axis:

$$K.E. = \frac{1}{2} I \omega^2$$

Where, I = moment of inertia, ω = angular velocity



EXAMPLE

A uniform disc of mass M and radius R rotates with angular speed ω about its central axis.

$$I_{\text{disc}} = \frac{1}{2} MR^2$$

$$K.E. = \frac{1}{2} \left(\frac{1}{2} MR^2 \right) \omega^2 = \frac{1}{4} MR^2 \omega^2$$

14. MOMENT OF INERTIA OF SOME COMMON BODIES

Body	Axis of Rotation	Moment of Inertia (I)
Thin Rod (length L , mass M)	Through centre, $\perp$ to length	$\frac{1}{12} ML^2$
Thin Rod (length L , mass M)	Through one end, $\perp$ to length	$\frac{1}{3} ML^2$
Solid Disc (radius R , mass M)	Through centre, $\perp$ to plane	$\frac{1}{2} MR^2$
Hollow Ring (radius R , mass M)	Through centre, $\perp$ to plane	MR^2
Solid Sphere (radius R , mass M)	Through centre	$\frac{2}{5} MR^2$
Thin Spherical Shell (radius R , mass M)	Through centre	$\frac{2}{3} MR^2$

IMPORTANT LINES

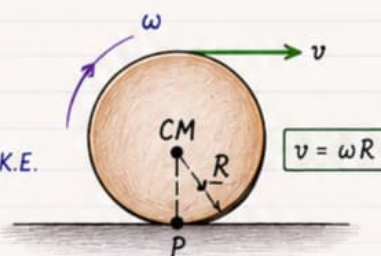
- Moment of inertia depends on the distribution of mass and axis of rotation.
- Greater the spread of mass from the axis, greater is I .
- For the same body, different axes give different I .
- Rotational K.E. is similar in form to linear K.E.

15. ROLLING MOTION

- When a body rolls on a surface without slipping, it has both translational and rotational motion.
- Total kinetic energy = Translational K.E. + Rotational K.E.
- For a body of mass M rolling with speed v :

$$K.E._{\text{total}} = \frac{1}{2} Mv^2 + \frac{1}{2} I \omega^2$$

- For rolling without slipping: $v = \omega R$



Example

A solid sphere (mass M , radius R) rolls without slipping with speed v .

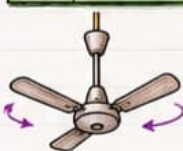
$$I_{\text{sphere}} = \frac{2}{5} MR^2$$

$$\omega = \frac{v}{R}$$

Total K.E.

$$= \frac{1}{2} Mv^2 + \frac{1}{2} \left(\frac{2}{5} MR^2 \right) \frac{v^2}{R^2} = \left(\frac{1}{2} + \frac{1}{5} \right) Mv^2 = \frac{7}{10} Mv^2$$

16. APPLICATIONS IN DAILY LIFE



Ceiling fan rotates about a fixed axis.



Wheel of a bicycle shows rolling motion.



Potter's wheel - rotational motion.



Spinning top (lattu) rotates about its axis.

KEY TAKEAWAYS

- ✓ System of particles is treated as a single entity for analysis.
- ✓ CM and total momentum help describe the system's motion.
- ✓ Rotational motion is described by θ , ω , α and related equations.
- ✓ Moment of inertia (I) is the rotational analogue of mass.
- ✓ Torque is the turning effect of a force.
- ✓ Rotational K.E. = $\frac{1}{2} I \omega^2$
- ✓ Rolling without slipping: $v = \omega R$

QUICK REVISION

- $R_{\text{cm}} = \frac{\sum m_i \vec{r}_i}{\sum m_i}$
- $\vec{P} = \sum \vec{p}_i$
- $I = \sum m_i r_i^2$
- $\vec{\tau} = \vec{r} \times \vec{F}$
- $\theta = \frac{s}{r}$ (small angle)
- $\omega = \frac{d\theta}{dt}$, $\alpha = \frac{d\omega}{dt}$
- $K.E._{\text{rot}} = \frac{1}{2} I \omega^2$
- $v = \omega R$ (rolling without slipping)



Remember!

- Practice diagrams → build strong concepts.
- Understand meanings → Score in NEET.
- NCERT lines → High weightage.

